

Notes: Cunha, Heckman, and Schennach (Econometrica, 2010)

Estimating the Technology of Cognitive and Noncognitive Skill Formation

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1 Big picture

- **Question.** How are children’s cognitive and noncognitive skills produced over the life cycle, how substitutable are early and late investments, and what do those substitution patterns imply for remediation policy?
- **Setting.** The paper combines a dynamic model of skill formation with rich child-development data from the Children of the NLSY79. Skills and parental investments are latent, so the empirical problem is to recover them from many noisy proxies.
- **Core challenge.** Child skills, parental investments, and even parental traits are not directly observed. Test scores are arbitrarily scaled, measurements are noisy, and parental investments are endogenous because parents react to what they know about the child.
- **Main conceptual contribution.** The paper develops a nonlinear dynamic factor framework that identifies and estimates a *multistage* skill-production technology with measurement error, endogenous inputs, and interpretable outcome scales.
- **Main economic mechanism.** Current skills raise future skills through *self-productivity*, and current skills can also raise the productivity of investment through *dynamic complementarity*. These two forces determine whether it is efficient to invest early or remediate later.
- **Main empirical result.** Cognitive-skill production becomes much less substitutable in later childhood, while noncognitive-skill production remains roughly equally substitutable across stages. In plain terms, cognitive deficits are much harder to repair late than early, whereas noncognitive deficits remain more remediable.
- **Cross-skill result.** Noncognitive skills help produce cognitive skills early in life, but the reverse channel from cognitive to noncognitive skills is weak or absent in the preferred specifications.
- **Measurement result.** Measurement error is economically large. Ignoring it materially distorts the estimated technology and can make it look as if early investments matter little for cognition.
- **Policy-simulation result.** When the planner chooses early and late investments to maximize schooling or reduce crime, the optimal policy generally tilts investment toward the *early* period for disadvantaged children.
- **Takeaway.** The paper gives a structural foundation for the “invest early” view of child policy, but with an important refinement: later interventions can still be productive when they operate through noncognitive skills.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper needs repeated measures of child skills, family investments, parental traits, and later-life outcomes.
- The data come from the **Children of the NLSY79** (CNLSY79), which repeatedly assesses children at two-year intervals and contains rich information on home inputs, child test scores, child behavior, and parental characteristics.

- The estimation sample contains **2,207 firstborn white children**. The firstborn restriction simplifies the dynamic family-investment problem, and the racial restriction is used to work with a more homogeneous sample for this paper’s baseline estimation.
- Assessments begin in 1986 and continue biennially. In the empirical implementation, one model period equals **two calendar years**.

2.2 Timing and stages of development

- The model has **eight childhood periods**, each spanning two years.
- Period 1 corresponds to birth, period 2 to ages 1–2, period 3 to ages 3–4, and so on up to ages 13–14.
- These eight periods are grouped into **two stages**:
 - **Stage 1**: birth through ages 5–6,
 - **Stage 2**: ages 5–6 through ages 13–14.
- The stage structure lets the technology be more flexible than a one-period childhood model, while still remaining tractable enough for estimation.

2.3 Latent objects the paper tries to recover

For child h in period t , the core latent objects are:

- cognitive skill $\theta_{C,t}$,
- noncognitive skill $\theta_{N,t}$,
- parental investment I_t ,
- parental cognitive skill $\theta_{C,P}$,
- parental noncognitive skill $\theta_{N,P}$,
- and shocks or unobserved inputs $\eta_{k,t}$ for $k \in \{C, N\}$.

Adult outcomes are produced by the skill stocks at the start of adulthood, denoted $\theta_{C,T+1}$ and $\theta_{N,T+1}$.

Interpretation. The paper is not estimating how one observed test score causes another. It is estimating how latent skills evolve and how noisy measurements map into those skills.

2.4 Measures used for the latent factors

Child cognitive skill measures.

- At very early ages: gestation length, birth weight, motor-social development.
- Ages 1–4: body-parts and memory tasks, motor-social development, PPVT.
- Ages 5–14: PIAT mathematics, reading recognition, and reading comprehension.

Child noncognitive skill measures.

- Early temperament-type measures: difficulty, friendliness, compliance, insecurity, sociability.
- Later behavior scales: Behavior Problem Index antisocial, anxiety, headstrong, hyperactive, conflict.

Parental skill measures.

- Maternal cognitive skill: ASVAB components such as arithmetic reasoning, word knowledge, paragraph comprehension, numerical operations, coding speed, and mathematical knowledge.
- Maternal noncognitive skill: self-esteem items and locus-of-control items.

Parental investment measures.

- Very early inputs: books, reading to the child, outings, toys, eating with parents, mother calling from work.
- Later home investments: books, museum visits, musical shows, special lessons, praise, encouragement, newspapers, instruments, magazines.

Important modeling simplification. The paper cannot empirically distinguish investment aimed specifically at cognition from investment aimed specifically at noncognition, so it imposes

$$I_{C,t} = I_{N,t} = I_t.$$

Interpretation. The empirical investment factor should be read as a common parental-input bundle that can affect both skill domains differently.

2.5 Measurement system

The main linear measurement equations are

$$Z_{1,k,t,j} = \mu_{1,k,t,j} + \alpha_{1,k,t,j} \ln \theta_{k,t} + \varepsilon_{1,k,t,j}, \quad (1)$$

$$Z_{2,k,t,j} = \mu_{2,k,t,j} + \alpha_{2,k,t,j} \ln I_t + \varepsilon_{2,k,t,j}, \quad (2)$$

$$Z_{3,k,1,j} = \mu_{3,k,1,j} + \alpha_{3,k,1,j} \ln \theta_{k,P} + \varepsilon_{3,k,1,j}. \quad (3)$$

The paper works with *log latent factors* in the measurement system so that the underlying factors themselves remain nonnegative when they enter the production technology.

Interpretation. Each observed test score or survey response is treated as a noisy loading on an underlying latent skill or latent investment factor.

2.6 Anchoring skills on an interpretable scale

A central problem in this literature is that test scores are measured on arbitrary scales. The paper solves this by **anchoring** latent skills in later outcomes that have an interpretable scale.

The main reported anchor is:

- completed years of schooling by age 19.

Interpretation. After anchoring, movements in latent skills can be interpreted in terms of their consequences for a meaningful adult outcome instead of in units of a raw psychometric score.

2.7 Descriptive facts about measurement error

The paper directly estimates how much of each observed measure is signal versus noise.

- Early cognitive proxies are often noisy. For example, motor-social development at birth has less than 5% signal.
- Some later cognitive tests are very informative. PIAT reading recognition at ages 5–6 has about **95.8%** signal.
- Noncognitive measures are, on average, noisier than cognitive measures. The paper reports that **less than 40%** of observed variance in noncognitive measures reflects true latent-skill variation.
- An extreme example is *sociability* at ages 3–4, where almost none of the observed variance is signal.
- Investment measures are noisier still. Many home-input measures have signal shares well below 25%, and several are below 10%.

Interpretation. This is why the paper insists on a factor-model approach. Using raw proxies directly in a production function would mechanically attenuate or distort the technology estimates.

3 Models and empirical specifications (all equations + interpretation)

3.1 General technology of skill formation

The most general production technology is

$$\theta_{k,t+1} = f_{k,s}(\theta_t, I_{k,t}, \theta_P, \eta_{k,t}), \quad k \in \{C, N\}. \quad (4)$$

Current skills can affect future skills in two ways:

- **Self-productivity:** today’s cognitive or noncognitive skill directly raises tomorrow’s same-domain skill.
- **Cross-productivity:** one skill today can help produce the other skill tomorrow.

Dynamic complementarity means current skills raise the productivity of investment:

$$\frac{\partial^2 f_{k,s}}{\partial I_{k,t} \partial \theta_{\ell,t}} > 0.$$

Interpretation. A more skilled child learns more from a given investment. That is the key force behind the possibility that early deficits become harder to repair later.

3.2 CES parameterization used in estimation

The estimated technology is a CES system. For cognitive skills:

$$\theta_{C,t+1} = \left[\gamma_{s,C,1} \theta_{C,t}^{\phi_{s,C}} + \gamma_{s,C,2} \theta_{N,t}^{\phi_{s,C}} + \gamma_{s,C,3} I_t^{\phi_{s,C}} + \gamma_{s,C,4} \theta_{C,P}^{\phi_{s,C}} + \gamma_{s,C,5} \theta_{N,P}^{\phi_{s,C}} \right]^{1/\phi_{s,C}}, \quad (5)$$

and similarly for noncognitive skills:

$$\theta_{N,t+1} = \left[\gamma_{s,N,1} \theta_{C,t}^{\phi_{s,N}} + \gamma_{s,N,2} \theta_{N,t}^{\phi_{s,N}} + \gamma_{s,N,3} I_t^{\phi_{s,N}} + \gamma_{s,N,4} \theta_{C,P}^{\phi_{s,N}} + \gamma_{s,N,5} \theta_{N,P}^{\phi_{s,N}} \right]^{1/\phi_{s,N}}. \quad (6)$$

The share parameters satisfy

$$\sum_{\ell=1}^5 \gamma_{s,k,\ell} = 1, \quad \gamma_{s,k,\ell} \geq 0.$$

The elasticity of substitution for skill k in stage s is

$$\sigma_{s,k} = \frac{1}{1 - \phi_{s,k}}.$$

Interpretation.

- Large $\sigma_{s,k}$ means current investment can easily compensate for low inherited skill.
- Small $\sigma_{s,k}$ means low inherited skill is hard to offset later.

3.3 Adult outcome equation and anchoring

Adult outcome Q_j is also written in CES form:

$$Q_j = \left[\rho_j \theta_{C,T+1}^{\phi_{Q_j}} + (1 - \rho_j) \theta_{N,T+1}^{\phi_{Q_j}} \right]^{1/\phi_{Q_j}}. \quad (7)$$

Interpretation. Different adult outcomes load differently on cognition and noncognition. Schooling depends more on cognition; crime is more noncognitive-intensive.

3.4 Two-period intuition for early versus late investment

To build intuition, the paper collapses the model to two childhood periods and one adult outcome. In that special case, optimal timing satisfies

$$\log\left(\frac{I_1}{I_2}\right) = \frac{1}{1 - \phi} \left[\log\left(\frac{\tau_1}{\tau_2}\right) - \log(1 + r) \right]. \quad (8)$$

Interpretation.

- If early investment is relatively more productive (τ_1/τ_2 high), optimal investment tilts toward period 1.
- If inputs are highly complementary (low ϕ , low elasticity), the planner wants a more balanced sequence of early and late investment.
- Figure 1 in the paper visualizes this timing logic.

3.5 Identification logic in the measurement model

The paper first identifies the distribution of the latent variables from repeated noisy measurements. The idea is:

- the *signal* is common across measurements,
- the *noise* is not.

Two central results structure the argument:

Theorem 1. With two sets of repeated measurements and weak orthogonality conditions on the measurement errors, the joint distribution of the latent factors can be identified.

Theorem 2. In a more general nonlinear measurement system, conditional independence plus completeness conditions identify the distribution of the latent variables.

Interpretation. The paper does not need to observe “true” skills. It only needs enough independent proxy variation to back out the common latent structure.

3.6 Why anchoring matters

Without anchoring, latent skill levels are identified only up to monotone transformations. The paper uses adult outcomes to fix a meaningful scale.

Interpretation. This is a major conceptual improvement over using test-score levels as if they were cardinal output measures. It lets the researcher say how a change in latent skill maps into schooling, crime, or other adult outcomes.

3.7 Baseline empirical specification

The paper estimates several versions of the technology. The baseline specification:

- corrects for measurement error,
- assumes the latent factors are normally distributed,
- anchors the latent scale in completed years of schooling by age 19,
- and *does not* yet control for unobserved heterogeneity π .

The technology is augmented with shocks:

$$\theta_{k,t+1} = \left[\gamma_{s,k,1} \theta_{C,t}^{\phi_{s,k}} + \gamma_{s,k,2} \theta_{N,t}^{\phi_{s,k}} + \gamma_{s,k,3} I_t^{\phi_{s,k}} + \gamma_{s,k,4} \theta_{C,P}^{\phi_{s,k}} + \gamma_{s,k,5} \theta_{N,P}^{\phi_{s,k}} \right]^{1/\phi_{s,k}} e^{\eta_{k,t+1}}. \quad (9)$$

3.8 Accounting for endogeneity of investments

The paper then allows for richer endogeneity by introducing unobserved heterogeneity and an explicit investment equation. The more general specification includes

$$I_t = k_C \theta_{C,t} + k_N \theta_{N,t} + k_{C,P} \theta_{C,P} + k_{N,P} \theta_{N,P} + k_y y_t + \pi_t, \quad (10)$$

with income and heterogeneity processes

$$\ln y_t = \rho_y \ln y_{t-1} + \nu_{y,t}, \quad (11)$$

$$\pi_t = \rho_\pi \pi_{t-1} + \nu_{\pi,t}. \quad (12)$$

Interpretation. Parents do not choose investments randomly. They respond to observed and unobserved characteristics of the child and family. This extension tries to make the technology estimates robust to that behavioral response.

3.9 Estimation method

Although the paper proves nonparametric identification, it estimates the model parametrically by maximum likelihood using nonlinear filtering.

Why parametric estimation?

- a fully nonparametric estimator would be too data-hungry in this high-dimensional setting;
- the dynamic factor structure with many latent states is computationally demanding;
- the parametric form delivers a tractable way to bring the identification theory to the data.

3.10 Policy simulation problem

After estimating the technology, the paper solves a planner’s problem. One version maximizes average schooling subject to a fixed budget:

$$\max \bar{S} = \frac{1}{H} \sum_{h=1}^H S(\theta_{C,3,h}, \theta_{N,3,h}, \pi_h) \quad (13)$$

$$\text{s.t. } \sum_{h=1}^H (I_{1,h} + I_{2,h}) = 2H, \quad (14)$$

with the technology constraining how skills evolve.

The paper also studies an analogous problem that minimizes aggregate crime.

Interpretation. These are not estimates of what actual parents do. They are policy counterfactuals asking: *if a planner could allocate resources optimally, when should disadvantaged children be helped?*

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the technology

The paper has no conventional external instrument. Identification comes from:

- many repeated measurements on the same latent constructs,

- dynamic restrictions linking skill stocks across childhood stages,
- outcome equations that anchor latent skills to interpretable adult outcomes,
- and additional structure for parental investment decisions when endogeneity is modeled explicitly.

Interpretation. Credibility here comes from the *structure* of the model and the richness of the measurement system, not from an exogenous shock like a policy reform or geographic instrument.

4.2 Why correcting measurement error is essential

This is one of the most important lessons of the paper. When the authors ignore measurement error and use even the “best” observed proxies directly:

- the first-stage investment effect on cognitive skill becomes small and imprecise,
- the second-stage investment effect on cognition becomes spuriously strong,
- and the model starts to produce strong cross-effects that do not survive once measurement error is handled properly.

Economic implication. A naive regression on raw child-development proxies could easily lead the researcher to the wrong policy conclusion about when investment matters most.

4.3 What the baseline estimates say

Baseline model with measurement error corrected (Table I).

For **cognitive skill formation**:

- self-productivity rises sharply from $\gamma_{1,C,1} = 0.487$ to $\gamma_{2,C,1} = 0.902$,
- the investment share drops from $\gamma_{1,C,3} = 0.231$ to $\gamma_{2,C,3} = 0.020$,
- the elasticity of substitution falls from about 2.57 in stage 1 to about 0.42 in stage 2.

For **noncognitive skill formation**:

- self-productivity rises from $\gamma_{1,N,2} = 0.649$ to $\gamma_{2,N,2} = 0.868$,
- the investment share falls more modestly from $\gamma_{1,N,3} = 0.146$ to $\gamma_{2,N,3} = 0.055$,
- the elasticity of substitution remains about constant: roughly 0.60 in both stages.

Interpretation.

- Skills become more persistent over time.
- Cognition becomes much harder to remediate late.
- Noncognitive skill remains much more malleable later in childhood.

4.4 How heterogeneity changes the estimates

Once the paper allows for time-invariant unobserved heterogeneity (Table IV), the main pattern survives and becomes even sharper.

Cognitive skill:

- investment share: 0.161 in stage 1 versus 0.044 in stage 2,
- elasticity: about 1.46 in stage 1 versus 0.45 in stage 2.

Noncognitive skill:

- investment share: 0.065 in stage 1 versus 0.051 in stage 2,
- elasticity: about 0.62 in stage 1 versus 0.65 in stage 2.

Interpretation. The main policy message is not an artifact of the simplest model. Even after allowing family inputs to be correlated with unobserved child or family characteristics, the late-stage elasticity for cognition stays much lower than the early-stage elasticity.

4.5 More general endogeneity control

In the most general reported specification (Table V), where the investment equation is estimated jointly with the skill technology:

- the early cognitive investment share rises further to $\gamma_{1,C,3} = 0.261$,
- the late cognitive investment share stays low at $\gamma_{2,C,3} = 0.044$,
- the early cognitive elasticity rises to about 2.41, while the late cognitive elasticity remains about 0.45.

For noncognitive skill:

- early investment share rises to 0.209,
- late investment share rises to 0.104,
- the elasticity remains roughly stable across stages at about 0.68 versus 0.66.

Interpretation. Endogeneity matters quantitatively, but it reinforces rather than overturns the core result: cognitive skill is especially sensitive to *early* investment.

4.6 What the paper can and cannot distinguish

What it does well.

- It distinguishes latent skill formation from noisy measurements.
- It separates cognitive and noncognitive production technologies.
- It delivers interpretable substitution parameters and policy counterfactuals.

What it does not do.

- It does not estimate a full general-equilibrium model of parental behavior under policy interventions.
- It does not model adult post-childhood investment.
- It cannot separately identify cognitive-specific and noncognitive-specific investments in the data, so it uses a common investment factor.
- It excludes schooling inputs from the main specification because those measures are relatively weak and noisy in the CNLSY.
- The main sample is restricted to firstborn white children, so external validity should be considered with care.

4.7 Relation to the literature

- Relative to **Cunha and Heckman (2008)**, this paper moves from a linear technology to a nonlinear CES technology and directly estimates substitution elasticities.
- Relative to **Todd and Wolpin**, it allows many noisy proxies and treats child skills as latent rather than as observed test scores.
- Relative to **Carneiro, Hansen, and Heckman**, it generalizes the nonlinear factor-analysis approach and emphasizes anchoring.
- Relative to **Olley and Pakes**-style proxy methods, it studies endogenous inputs when the proxies are imperfect and measured with error.

5 Tables and figures

5.1 Figure 1: intuition for early versus late investment

Read:

- Figure 1 plots the optimal ratio of early to late investment against the relative productivity of early investment.
- When early investment is relatively more productive, the planner tilts investment toward the first stage.
- The shape of the response depends strongly on complementarity. Under perfect complements, the ratio stays near one; under stronger substitutability, the ratio moves more sharply.

Economic message. The timing problem depends on both *productivity differences across stages* and *substitutability across inputs*. This is the paper's basic conceptual map.

5.2 Table I: baseline technology with measurement error correction

Read:

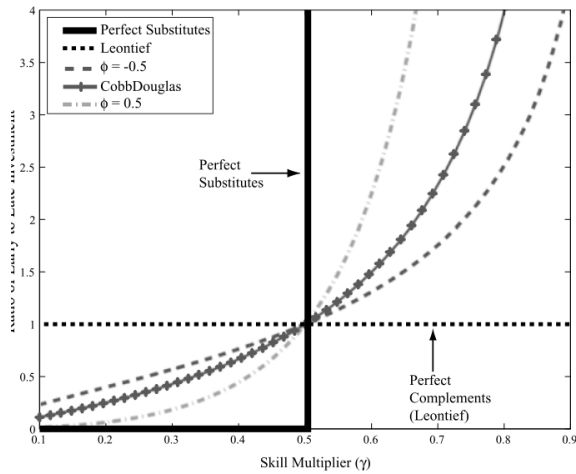


FIGURE 1.—Ratio of early to late investment in human capital as a function of the ratio of first period to second period investment productivity for different values of the complementarity parameter; assumes $r = 0$. Source: Cunha and Heckman (2007).

- For cognition, self-productivity is stronger in stage 2, but the investment share is much larger in stage 1.
- The implied elasticity of substitution for cognition collapses from about 2.57 to about 0.42.
- For noncognition, self-productivity also rises with age, but the elasticity of substitution stays around 0.60 in both stages.

Economic message. The basic “invest early for cognition” result already appears in the simplest measurement-error-corrected specification.

5.3 Tables IIA and IIB: signal versus noise in the observed proxies

TABLE IIA
PERCENTAGE OF TOTAL VARIANCE IN MEASUREMENTS DUE TO SIGNAL AND NOISE

	%Signal	%Noise		%Signal	%Noise
<i>Measurement of Child's Cognitive Skills</i>					
Gestation Length	0.501	0.499	<i>Measurement of Child's Noncognitive Skills</i>		
Weight at Birth	0.557	0.443	Difficulty at Birth	0.151	0.849
Motor-Social Development at Birth	0.045	0.955	Friendliness at Birth	0.165	0.835
Body Parts at Ages 1-2	0.275	0.725	Compliance at Ages 1-2	0.232	0.768
Memory for Locations at Ages 1-2	0.306	0.692	Insecure at Ages 1-2	0.080	0.920
Motor-Social Development at Ages 3-4	0.160	0.840	Sociability at Ages 1-2	0.075	0.925
Picture Vocabulary at Ages 3-4	0.410	0.590	Difficulty at Ages 1-2	0.382	0.618
Picture Vocabulary at Ages 5-6	0.431	0.569	Friendliness at Ages 1-2	0.189	0.811
PIAT Reading Recognition at Ages 5-6	0.225	0.775	Compliance at Ages 3-4	0.133	0.867
PIAT Mathematics at Ages 5-6	0.314	0.686	Insecure at Ages 3-4	0.122	0.878
PIAT Reading Comprehension at Ages 5-6	0.958	0.042	Sociability at Ages 3-4	0.008	0.992
PIAT Mathematics at Ages 7-8	0.465	0.535	Behavior Problem Index Antisocial at Ages 3-4	0.405	0.595
PIAT Reading Recognition at Ages 7-8	0.869	0.131	Behavior Problem Index Anxiety at Ages 3-4	0.427	0.573
PIAT Reading Comprehension at Ages 7-8	0.797	0.203	Behavior Problem Index Headstrong at Ages 3-4	0.518	0.482
PIAT Mathematics at Ages 9-10	0.492	0.508	Behavior Problem Index Hyperactive at Ages 3-4	0.558	0.442
PIAT Reading Recognition at Ages 9-10	0.817	0.183	Behavior Problem Index Conflict at Ages 3-4	0.336	0.664
PIAT Reading Comprehension at Ages 9-10	0.666	0.334	Behavior Problem Index Antisocial at Ages 5-6	0.435	0.565
PIAT Mathematics at Ages 11-12	0.516	0.484	Behavior Problem Index Anxiety at Ages 5-6	0.409	0.591
PIAT Reading Recognition at Ages 11-12	0.781	0.219	Behavior Problem Index Headstrong at Ages 5-6	0.611	0.389
PIAT Reading Comprehension at Ages 11-12	0.614	0.386	Behavior Problem Index Hyperactive at Ages 5-6	0.481	0.519
PIAT Mathematics at Ages 13-14	0.537	0.463	Behavior Problem Index Conflict at Ages 5-6	0.290	0.710
PIAT Reading Recognition at Ages 13-14	0.735	0.265	Behavior Problem Index Antisocial Ages 7-8	0.446	0.554
PIAT Reading Comprehension at Ages 13-14	0.549	0.451	Behavior Problem Index Anxiety Ages 7-8	0.475	0.525
			Behavior Problem Index Headstrong Ages 7-8	0.605	0.395
			Behavior Problem Index Hyperactive Ages 7-8	0.497	0.503

(Continued)

TABLE I

USING THE FACTOR MODEL TO CORRECT FOR MEASUREMENT ERROR: LINEAR ANCHORING ON EDUCATIONAL ATTAINMENT (YEARS OF SCHOOLING); NO UNOBSERVED HETEROGENEITY (π), FACTORS NORMALLY DISTRIBUTED*

	First Stage Parameters	Second Stage Parameters
<i>The Technology of Cognitive Skill Formation</i>		
Current Period Cognitive Skills (Self-Productivity)	$\gamma_{1,C,1}$ 0.487 (0.030)	$\gamma_{2,C,1}$ 0.902 (0.014)
Current Period Noncognitive Skills (Cross-Productivity)	$\gamma_{1,C,2}$ 0.083 (0.026)	$\gamma_{2,C,2}$ 0.011 (0.005)
Current Period Investments	$\gamma_{1,C,3}$ 0.231 (0.024)	$\gamma_{2,C,3}$ 0.020 (0.006)
Parental Cognitive Skills	$\gamma_{1,C,4}$ 0.050 (0.013)	$\gamma_{2,C,4}$ 0.047 (0.008)
Parental Noncognitive Skills	$\gamma_{1,C,5}$ 0.148 (0.030)	$\gamma_{2,C,5}$ 0.020 (0.010)
Complementarity Parameter	$\phi_{1,C}$ 0.611 (0.240)	$\phi_{2,C}$ -1.373 (0.168)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,C})$ 2.569	$1/(1 - \phi_{2,C})$ 0.421
Variance of Shocks $\eta_{C,i}$	$\delta_{1,C}^2$ 0.165 (0.007)	$\delta_{2,C}^2$ 0.097 (0.003)
<i>The Technology of Noncognitive Skill Formation</i>		
Current Period Cognitive Skills (Cross-Productivity)	$\gamma_{1,N,1}$ 0.000 (0.025)	$\gamma_{2,N,1}$ 0.008 (0.010)
Current Period Noncognitive Skills (Self-Productivity)	$\gamma_{1,N,2}$ 0.649 (0.034)	$\gamma_{2,N,2}$ 0.868 (0.011)
Current Period Investments	$\gamma_{1,N,3}$ 0.146 (0.027)	$\gamma_{2,N,3}$ 0.055 (0.013)
Parental Cognitive Skills	$\gamma_{1,N,4}$ 0.022 (0.011)	$\gamma_{2,N,4}$ 0.000 (0.007)
Parental Noncognitive Skills	$\gamma_{1,N,5}$ 0.183 (0.031)	$\gamma_{2,N,5}$ 0.069 (0.017)
Complementarity Parameter	$\phi_{1,N}$ -0.674 (0.324)	$\phi_{2,N}$ -0.695 (0.274)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,N})$ 0.597	$1/(1 - \phi_{2,N})$ 0.590
Variance of Shocks $\eta_{N,i}$	$\delta_{1,N}^2$ 0.189 (0.012)	$\delta_{2,N}^2$ 0.103 (0.004)

*Standard errors in parentheses.

	%Signal	%Noise		%Signal	%Noise
<i>Measurements of Parental Investments</i>					
How Often Child Goes on Outings During Year of Birth	0.329	0.671	Child Has Musical Instruments Ages 7-8	0.022	0.978
Number of Books Child Has During Year of Birth	0.289	0.791	Family Subscribes to Daily Newspapers Ages 7-8	0.023	0.977
How Often Mom Reads to Child During Year of Birth	0.484	0.516	Child Has Special Lessons Ages 7-8	0.018	0.982
Number of Soft Toys Child Has During Year of Birth	0.128	0.872	How Often Child Goes to Musical Shows Ages 7-8	0.266	0.734
Number of Push/Pull Toys Child Has During Year of Birth	0.019	0.981	How Often Child Attends Family Gatherings Ages 7-8	0.125	0.875
How Often Child Eats With Mom/Dad During Year of Birth	0.511	0.489	How Often Child Is Praised Ages 7-8	0.046	0.954
How Often Mom Calls From Work During Year of Birth	0.119	0.881	How Often Child Gets Positive Encouragement Ages 7-8	0.053	0.947
How Often Child Goes on Outings at Ages 1-2	0.148	0.852	Number of Books Child Has Ages 9-10	0.013	0.987
Number of Books Child Has Ages 1-2	0.055	0.945	Mom Reads to Child Ages 9-10	0.137	0.863
How Often Mom Reads to Child Ages 1-2	0.186	0.814	Eat With Mom/Dad Ages 9-10	0.162	0.838
Number of Soft Toys Child Has Ages 1-2	0.240	0.760	How Often Child Goes to Museum Ages 9-10	0.219	0.781
Number of Push/Pull Toys Child Has Ages 1-2	0.046	0.954	Child Has Musical Instruments Ages 9-10	0.019	0.981
How Often Child Eats With Mom/Dad Ages 1-2	0.194	0.806	Family Subscribes to Daily Newspapers Ages 9-10	0.019	0.981
Mom Calls From Work Ages 1-2	0.070	0.930	Child Has Special Lessons Ages 9-10	0.015	0.985
How Often Child Goes on Outings Ages 3-4	0.123	0.877	How Often Child Goes to Musical Shows Ages 9-10	0.242	0.758
Number of Books Child Has Ages 3-4	0.012	0.988	How Often Child Attends Family Gatherings Ages 9-10	0.115	0.885
How Often Mom Reads to Child Ages 3-4	0.088	0.912	How Often Child Is Praised Ages 9-10	0.036	0.964
How Often Child Eats With Mom/Dad Ages 3-4	0.170	0.830	How Often Child Gets Positive Encouragement Ages 9-10	0.041	0.959
Number of Magazines at Home Ages 3-4	0.193	0.807	Number of Books Child Has Ages 11-12	0.016	0.984

(Continued)

Read:

- Table IIA shows that some cognitive tests are highly informative, but many early measures and many noncognitive measures are very noisy.
- Table IIB shows that parental-investment proxies are even noisier than skill proxies.
- Later cognitive tests such as PIAT reading recognition contain much more signal than early-life developmental measures.

Economic message. A raw-input, raw-test-score production function would be severely contaminated by measurement error.

5.4 Tables III, IV, and V: why measurement error and endogeneity matter

TABLE III THE TECHNOLOGY FOR COGNITIVE AND NONCOGNITIVE SKILL FORMATION: NOT CORRECTING FOR MEASUREMENT ERROR; LINEAR ANCHORING ON EDUCATIONAL ATTAINMENT (YEARS OF SCHOOLING); NO UNOBSERVED HETEROGENEITY (η); FACTORS NORMALLY DISTRIBUTED ^a			
	First Stage Parameters		Second Stage Parameters
Panel A: Technology of Cognitive Skill Formation (Next Period Cognitive Skills)			
Current Period Cognitive Skills (Self-Productivity)	$\gamma_{1,C,1}$	0.403 (0.058)	$\gamma_{2,C,1}$ 0.657 (0.013)
Current Period Noncognitive Skills (Cross-Productivity)	$\gamma_{1,C,2}$	0.218 (0.105)	$\gamma_{2,C,2}$ 0.009 (0.005)
Current Period Investments	$\gamma_{1,C,3}$	0.167 (0.090)	$\gamma_{2,C,3}$ 0.167 (0.018)
Parental Cognitive Skills	$\gamma_{1,C,4}$	0.268 (0.078)	$\gamma_{2,C,4}$ 0.047 (0.009)
Parental Noncognitive Skills	$\gamma_{1,C,5}$	0.044 (0.050)	$\gamma_{2,C,5}$ 0.119 (0.150)
Complementarity Parameter	$\phi_{1,C}$	0.375 (0.294)	$\phi_{2,C}$ -0.827 (0.093)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,C})$	1.601	$1/(1 - \phi_{2,C})$ 0.547
Variance of Shocks $\eta_{C,t}$	$\hat{\sigma}_{1,C}^2$	0.941 (0.048)	$\hat{\sigma}_{2,C}^2$ 0.358 (0.006)
Panel B: Technology of Noncognitive Skill Formation (Next Period Noncognitive Skills)			
Current Period Cognitive Skills (Cross-Productivity)	$\gamma_{1,N,1}$	0.193 (0.095)	$\gamma_{2,N,1}$ 0.058 (0.014)
Current Period Noncognitive Skills (Self-Productivity)	$\gamma_{1,N,2}$	0.594 (0.090)	$\gamma_{2,N,2}$ 0.638 (0.020)
Current Period Investments	$\gamma_{1,N,3}$	0.099 (0.296)	$\gamma_{2,N,3}$ 0.239 (0.031)
Parental Cognitive Skills	$\gamma_{1,N,4}$	0.114 (0.055)	$\gamma_{2,N,4}$ 0.065 (0.015)
Parental Noncognitive Skills	$\gamma_{1,N,5}$	0.000 (0.821)	$\gamma_{2,N,5}$ 0.000 (0.203)
Complementarity Parameter	$\phi_{1,N}$	-0.723 (0.441)	$\phi_{2,N}$ -0.716 (0.127)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,N})$	0.580	$1/(1 - \phi_{2,N})$ 0.583
Variance of Shocks $\eta_{N,t}$	$\hat{\sigma}_{1,N}^2$	0.767 (0.076)	$\hat{\sigma}_{2,N}^2$ 0.597 (0.017)
*Standard errors in parentheses.			

TABLE IV THE TECHNOLOGY FOR COGNITIVE AND NONCOGNITIVE SKILL FORMATION: LINEAR ANCHORING ON EDUCATIONAL ATTAINMENT (YEARS OF SCHOOLING); ALLOWING FOR UNOBSERVED HETEROGENEITY (η); FACTORS NORMALLY DISTRIBUTED ^a			
	First Stage Parameters		Second Stage Parameters
Panel A: Technology of Cognitive Skill Formation (Next Period Cognitive Skills)			
Current Period Cognitive Skills (Self-Productivity)	$\gamma_{1,C,1}$	0.479 (0.026)	$\gamma_{2,C,1}$ 0.831 (0.011)
Current Period Noncognitive Skills (Cross-Productivity)	$\gamma_{1,C,2}$	0.070 (0.024)	$\gamma_{2,C,2}$ 0.001 (0.005)
Current Period Investments	$\gamma_{1,C,3}$	0.161 (0.015)	$\gamma_{2,C,3}$ 0.044 (0.006)
Parental Cognitive Skills	$\gamma_{1,C,4}$	0.031 (0.013)	$\gamma_{2,C,4}$ 0.073 (0.008)
Parental Noncognitive Skills	$\gamma_{1,C,5}$	0.258 (0.029)	$\gamma_{2,C,5}$ 0.051 (0.014)
Complementarity Parameter	$\phi_{1,C}$	0.313 (0.134)	$\phi_{2,C}$ -1.243 (0.125)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,C})$	1.457	$1/(1 - \phi_{2,C})$ 0.446
Variance of Shocks $\eta_{C,t}$	$\hat{\sigma}_{1,C}^2$	0.176 (0.007)	$\hat{\sigma}_{2,C}^2$ 0.087 (0.003)
Panel B: Technology of Noncognitive Skill Formation (Next Period Noncognitive Skills)			
Current Period Cognitive Skills (Cross-Productivity)	$\gamma_{1,N,1}$	0.000 (0.026)	$\gamma_{2,N,1}$ 0.000 (0.010)
Current Period Noncognitive Skills (Self-Productivity)	$\gamma_{1,N,2}$	0.585 (0.032)	$\gamma_{2,N,2}$ 0.816 (0.013)
Current Period Investments	$\gamma_{1,N,3}$	0.065 (0.021)	$\gamma_{2,N,3}$ 0.051 (0.006)
Parental Cognitive Skills	$\gamma_{1,N,4}$	0.017 (0.013)	$\gamma_{2,N,4}$ 0.000 (0.008)
Parental Noncognitive Skills	$\gamma_{1,N,5}$	0.333 (0.034)	$\gamma_{2,N,5}$ 0.133 (0.017)
Complementarity Parameter	$\phi_{1,N}$	-0.610 (0.215)	$\phi_{2,N}$ -0.551 (0.169)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,N})$	0.621	$1/(1 - \phi_{2,N})$ 0.645
Variance of Shocks $\eta_{N,t}$	$\hat{\sigma}_{1,N}^2$	0.222 (0.013)	$\hat{\sigma}_{2,N}^2$ 0.101 (0.004)
*Standard errors in parentheses.			

TABLE V THE TECHNOLOGY FOR COGNITIVE AND NONCOGNITIVE SKILL FORMATION ESTIMATED ALONG WITH INVESTMENT EQUATION WITH LINEAR ANCHORING ON EDUCATIONAL ATTAINMENT (YEARS OF SCHOOLING); FACTORS NORMALLY DISTRIBUTED ^a			
	First Stage Parameters		Second Stage Parameters
Panel A: Technology of Cognitive Skill Formation (Next Period Cognitive Skills)			
Current Period Cognitive Skills (Self-Productivity)	$\gamma_{1,C,1}$	0.485 (0.031)	$\gamma_{2,C,1}$ 0.884 (0.015)
Current Period Noncognitive Skills (Cross-Productivity)	$\gamma_{1,C,2}$	0.062 (0.026)	$\gamma_{2,C,2}$ 0.011 (0.005)
Current Period Investments	$\gamma_{1,C,3}$	0.261 (0.026)	$\gamma_{2,C,3}$ 0.044 (0.011)
Parental Cognitive Skills	$\gamma_{1,C,4}$	0.035 (0.015)	$\gamma_{2,C,4}$ 0.051 (0.008)
Parental Noncognitive Skills	$\gamma_{1,C,5}$	0.157 (0.033)	$\gamma_{2,C,5}$ 0.011 (0.012)
Complementarity Parameter	$\phi_{1,C}$	0.585 (0.225)	$\phi_{2,C}$ -1.229 (0.149)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,C})$	2.410	$1/(1 - \phi_{2,C})$ 0.450
Variance of Shocks $\eta_{C,t}$	$\hat{\sigma}_{1,C}^2$	0.165 (0.007)	$\hat{\sigma}_{2,C}^2$ 0.098 (0.005)
Panel B: Technology of Noncognitive Skill Formation (Next Period Noncognitive Skills)			
Current Period Cognitive Skills (Cross-Productivity)	$\gamma_{1,N,1}$	0.000 (0.028)	$\gamma_{2,N,1}$ 0.002 (0.011)
Current Period Noncognitive Skills (Self-Productivity)	$\gamma_{1,N,2}$	0.602 (0.034)	$\gamma_{2,N,2}$ 0.857 (0.011)
Current Period Investments	$\gamma_{1,N,3}$	0.209 (0.031)	$\gamma_{2,N,3}$ 0.104 (0.022)
Parental Cognitive Skills	$\gamma_{1,N,4}$	0.014 (0.013)	$\gamma_{2,N,4}$ 0.000 (0.008)
Parental Noncognitive Skills	$\gamma_{1,N,5}$	0.175 (0.033)	$\gamma_{2,N,5}$ 0.037 (0.021)
Complementarity Parameter	$\phi_{1,N}$	-0.464 (0.263)	$\phi_{2,N}$ -0.522 (0.214)
Implied Elasticity of Substitution	$1/(1 - \phi_{1,N})$	0.683	$1/(1 - \phi_{2,N})$ 0.615
Variance of Shocks $\eta_{N,t}$	$\hat{\sigma}_{1,N}^2$	0.205 (0.012)	$\hat{\sigma}_{2,N}^2$ 0.102 (0.003)
*Standard errors in parentheses.			

Read:

- Table III shows that ignoring measurement error pushes the estimated technology in the wrong direction, especially for the timing of cognitive investment.
- Table IV shows that allowing for unobserved heterogeneity leaves the qualitative pattern intact: early cognitive investment matters much more, and the late cognitive elasticity remains very low.
- Table V shows that jointly estimating the parental investment equation strengthens the estimated role of early investment even further.

Economic message. The paper's main result is robust to more realistic handling of endogeneity and is *not* an artifact of a knife-edge baseline specification.

5.5 Figures 2–6: optimal intervention profiles

Read:

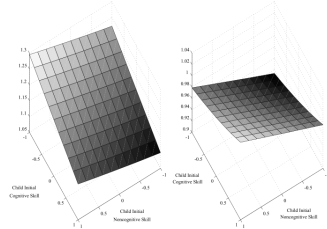


FIGURE 2.—Optimal early (left) and late (right) investments by child initial conditions of cognitive and noncognitive skills maximizing aggregate education (other endowments held at mean levels).

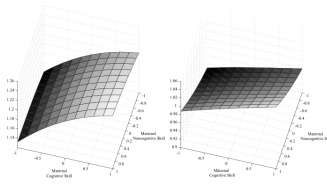


FIGURE 3.—Optimal early (left) and late (right) investments by maternal cognitive and noncognitive skills maximizing aggregate education (other endowments held at mean levels).

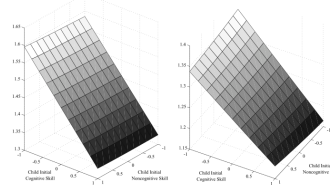


FIGURE 4.—Ratio of early to late investments by child initial conditions of cognitive and noncognitive skills maximizing aggregate education (left) and minimizing aggregate crime (right) (other endowments held at mean levels).

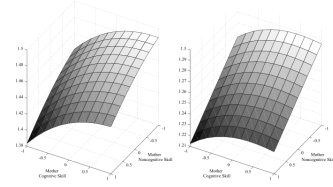


FIGURE 5.—Ratio of early to late investments by maternal cognitive and noncognitive skills maximizing aggregate education (left) and minimizing aggregate crime (right) (other endowments held at mean levels).

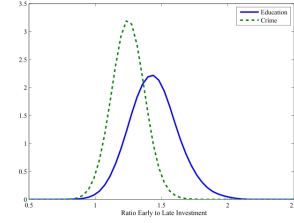


FIGURE 6.—Densities of ratio of early to late investments maximizing aggregate education versus minimizing aggregate crime.

- Figures 2 and 3 show the planner's optimal early and late investments across child and maternal endowments when the objective is to maximize schooling.
- Figures 4 and 5 show that the ratio of early to late investment declines with advantage: the more disadvantaged the child, the more strongly the planner wants to tilt resources toward the early period.
- Figure 6 shows that the density of the early-to-late investment ratio lies above one for most cases, with a stronger early tilt when the objective is education than when the objective is crime reduction.

Economic message. The estimated technology implies that equity and efficiency are not in conflict here: targeting disadvantaged children *early* is also the efficient policy.

6 Short summary

- The paper estimates a two-stage CES technology for cognitive and noncognitive skill formation using a dynamic factor model with many noisy proxies.
- Skills and investments are latent; adult outcomes are used to anchor the latent scale in interpretable units.
- Measurement error is large and cannot be ignored.
- The key result is that cognitive-skill production is much less substitutable in later childhood, while noncognitive-skill production remains roughly equally substitutable across stages.
- Early investment is therefore especially important for cognition, while later remediation is more promising when it works through noncognitive skills.
- Policy simulations imply that, for most disadvantaged children, the efficient allocation tilts resources toward early childhood rather than later remediation.

7 Conclusion

7.1 Core conclusion

The paper establishes that the technology of skill formation is dynamic, nonlinear, and domain specific. The main structural fact is that cognition and noncognition do *not* have the same substitution possibilities over the life cycle: cognitive deficits are much harder to repair late, while noncognitive traits remain more malleable.

7.2 Economic intuition

Two forces drive the result. First, skills are self-productive, so early advantages persist and compound. Second, early skills make later investment more productive. When the elasticity of substitution falls later in life, the cost of compensating for early cognitive disadvantage rises sharply. That is why the planner wants to move resources forward in time for disadvantaged children.

7.3 Takeaway

This paper is one of the clearest structural arguments for early-childhood investment in economics. Its most durable lesson is not simply “invest early,” but rather:

- invest early to build cognition,
- recognize that noncognitive skills remain an important remediation margin later on,
- and never estimate child skill technologies as if raw test scores and raw home-input proxies were error-free.